

```
/* -*- Mode:C++; c-file-style:"gnu"; indent-tabs-mode:nil; -*- */  
  
/*  
 * This program is free software; you can redistribute it and/or modify  
 * it under the terms of the GNU General Public License version 2 as  
 * published by the Free Software Foundation;  
 *  
 * This program is distributed in the hope that it will be useful,  
 * but WITHOUT ANY WARRANTY; without even the implied warranty of  
 * MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the  
 * GNU General Public License for more details.  
 *  
 * You should have received a copy of the GNU General Public License  
 * along with this program; if not, write to the Free Software  
 * Foundation, Inc., 59 Temple Place, Suite 330, Boston, MA 02111-1307 USA  
 */
```

```
#include "ns3/core-module.h"  
#include "ns3/point-to-point-module.h"  
#include "ns3/network-module.h"  
#include "ns3/applications-module.h"  
#include "ns3/mobility-module.h"  
#include "ns3/csma-module.h"  
#include "ns3/internet-module.h"  
#include "ns3/yans-wifi-helper.h"  
#include "ns3/ssid.h"  
#include "ns3/netanim-module.h"
```

```
// Default Network Topology  
  
//
```

```
// Wifi 10.1.3.0
//      AP
// * * * *
// | | | | 10.1.1.0
// n5 n6 n7 n0 ----- n1 n2 n3 n4
//      point-to-point | | | |
//      =====
//      LAN 10.1.2.0
```

```
using namespace ns3;
```

```
NS_LOG_COMPONENT_DEFINE ("ThirdScriptExample");
```

```
int
```

```
main (int argc, char *argv[])
```

```
{
```

```
    bool verbose = true;
```

```
    uint32_t nCsma = 3;
```

```
    uint32_t nWifi = 3;
```

```
    bool tracing = false;
```

```
    CommandLine cmd (__FILE__);
```

```
    cmd.AddValue ("nCsma", "Number of \"extra\" CSMA nodes/devices", nCsma);
```

```
    cmd.AddValue ("nWifi", "Number of wifi STA devices", nWifi);
```

```
    cmd.AddValue ("verbose", "Tell echo applications to log if true", verbose);
```

```
    cmd.AddValue ("tracing", "Enable pcap tracing", tracing);
```

```
    cmd.Parse (argc,argv);
```

```

// The underlying restriction of 18 is due to the grid position
// allocator's configuration; the grid layout will exceed the
// bounding box if more than 18 nodes are provided.
if (nWifi > 18)
{
    std::cout << "nWifi should be 18 or less; otherwise grid layout exceeds the bounding box" <<
std::endl;
    return 1;
}

if (verbose)
{
    LogComponentEnable ("UdpEchoClientApplication", LOG_LEVEL_INFO);
    LogComponentEnable ("UdpEchoServerApplication", LOG_LEVEL_INFO);
}

```

```

NodeContainer p2pNodes;

```

```

p2pNodes.Create (2);

```

```

PointToPointHelper pointToPoint;
pointToPoint.SetDeviceAttribute ("DataRate", StringValue ("5Mbps"));
pointToPoint.SetChannelAttribute ("Delay", StringValue ("2ms"));

```

```

NetDeviceContainer p2pDevices;
p2pDevices = pointToPoint.Install (p2pNodes);

```

```

NodeContainer csmaNodes;
csmaNodes.Add (p2pNodes.Get (1));
csmaNodes.Create (nCsmas);

```

```
CsmaHelper csma;  
csma.SetChannelAttribute ("DataRate", StringValue ("100Mbps"));  
csma.SetChannelAttribute ("Delay", TimeValue (NanoSeconds (6560)));
```

```
NetDeviceContainer csmaDevices;  
csmaDevices = csma.Install (csmaNodes);
```

```
NodeContainer wifiStaNodes;  
wifiStaNodes.Create (nWifi);  
NodeContainer wifiApNode = p2pNodes.Get (0);
```

```
YansWifiChannelHelper channel = YansWifiChannelHelper::Default ();  
YansWifiPhyHelper phy;  
phy.SetChannel (channel.Create ());
```

```
WifiHelper wifi;  
wifi.SetRemoteStationManager ("ns3::AarfWifiManager");
```

```
WifiMacHelper mac;  
Ssid ssid = Ssid ("ns-3-ssid");  
mac.SetType ("ns3::StaWifiMac",  
             "Ssid", SsidValue (ssid),  
             "ActiveProbing", BooleanValue (false));
```

```
NetDeviceContainer staDevices;  
staDevices = wifi.Install (phy, mac, wifiStaNodes);
```

```
mac.SetType ("ns3::ApWifiMac",
```

```
"Ssid", SsidValue (ssid));
```

```
NetDeviceContainer apDevices;
```

```
apDevices = wifi.Install (phy, mac, wifiApNode);
```

```
MobilityHelper mobility;
```

```
mobility.SetPositionAllocator ("ns3::GridPositionAllocator",
```

```
    "MinX", DoubleValue (0.0),
```

```
    "MinY", DoubleValue (0.0),
```

```
    "DeltaX", DoubleValue (5.0),
```

```
    "DeltaY", DoubleValue (10.0),
```

```
    "GridWidth", UIntegerValue (3),
```

```
    "LayoutType", StringValue ("RowFirst"));
```

```
mobility.SetMobilityModel ("ns3::RandomWalk2dMobilityModel",
```

```
    "Bounds", RectangleValue (Rectangle (-50, 50, -50, 50)));
```

```
mobility.Install (wifiStaNodes);
```

```
mobility.SetMobilityModel ("ns3::ConstantPositionMobilityModel");
```

```
mobility.Install (wifiApNode);
```

```
InternetStackHelper stack;
```

```
stack.Install (csmaNodes);
```

```
stack.Install (wifiApNode);
```

```
stack.Install (wifiStaNodes);
```

```
Ipv4AddressHelper address;
```

```
address.SetBase ("10.1.1.0", "255.255.255.0");  
Ipv4InterfaceContainer p2pInterfaces;  
p2pInterfaces = address.Assign (p2pDevices);
```

```
address.SetBase ("10.1.2.0", "255.255.255.0");  
Ipv4InterfaceContainer csmaInterfaces;  
csmaInterfaces = address.Assign (csmaDevices);
```

```
address.SetBase ("10.1.3.0", "255.255.255.0");  
address.Assign (staDevices);  
address.Assign (apDevices);
```

```
UdpEchoServerHelper echoServer (9);
```

```
ApplicationContainer serverApps = echoServer.Install (csmaNodes.Get (nCsm));  
serverApps.Start (Seconds (1.0));  
serverApps.Stop (Seconds (10.0));
```

```
UdpEchoClientHelper echoClient (csmaInterfaces.GetAddress (nCsm), 9);  
echoClient.SetAttribute ("MaxPackets", UIntegerValue (1));  
echoClient.SetAttribute ("Interval", TimeValue (Seconds (1.0)));  
echoClient.SetAttribute ("PacketSize", UIntegerValue (1024));
```

```
ApplicationContainer clientApps =  
    echoClient.Install (wifiStaNodes.Get (nWifi - 1));  
clientApps.Start (Seconds (2.0));  
clientApps.Stop (Seconds (10.0));
```

```
Ipv4GlobalRoutingHelper::PopulateRoutingTables ();
```

```
Simulator::Stop (Seconds (10.0));
```

```
if (tracing)
```

```
{
```

```
    phy.SetPcapDataLinkType (WifiPhyHelper::DLT_IEEE802_11_RADIO);
```

```
    pointToPoint.EnablePcapAll ("third");
```

```
    phy.EnablePcap ("third", apDevices.Get (0));
```

```
    csma.EnablePcap ("third", csmaDevices.Get (0), true);
```

```
}
```

```
// Create Animation Interface
```

```
AnimationInterface anim("hybridanim.xml");
```

```
// Set constant position for nodes in the animation
```

```
anim.SetConstantPosition(p2pNodes.Get(0), 10.0, 10.0); // Position for node 0
```

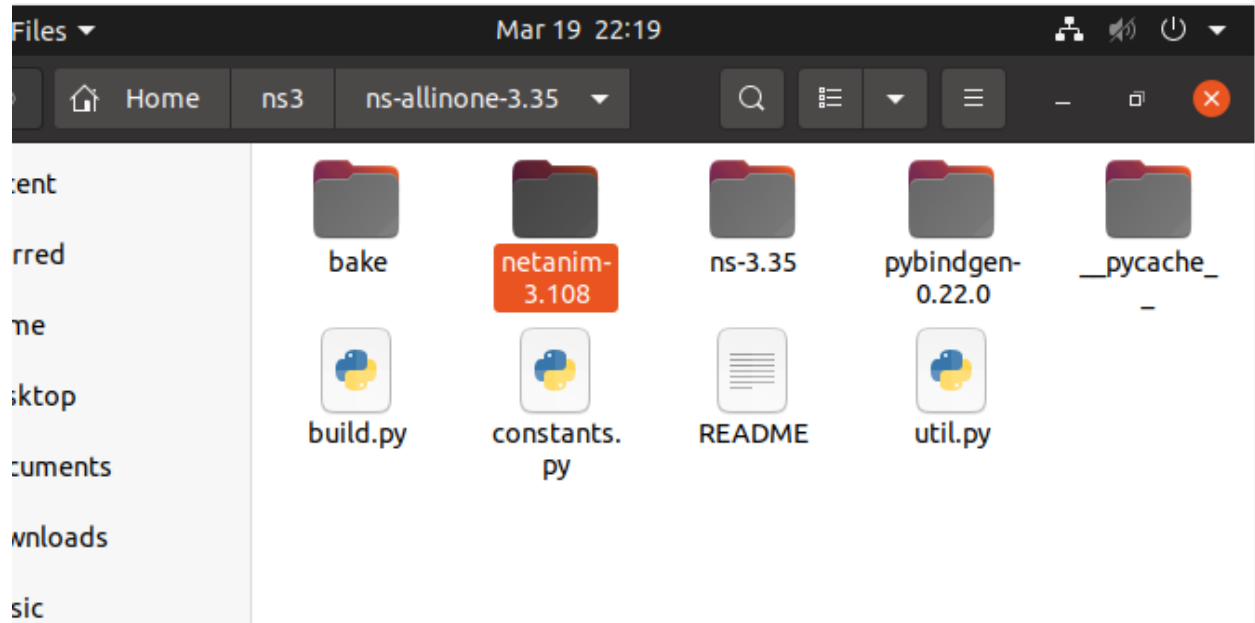
```
anim.SetConstantPosition(p2pNodes.Get(1), 20.0, 20.0); // Position for node 1
```

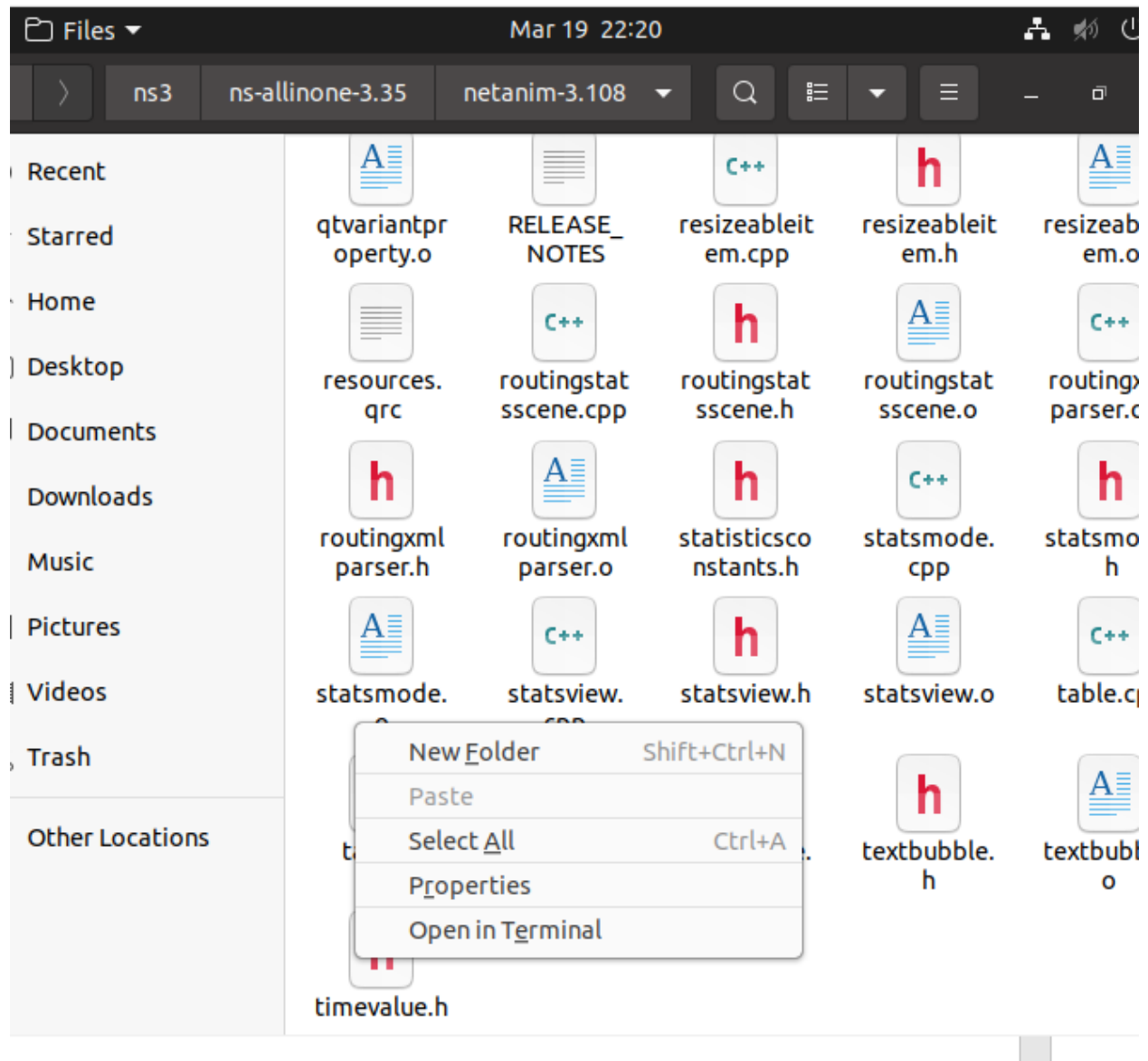
```
Simulator::Run ();
```

```
Simulator::Destroy ();
```

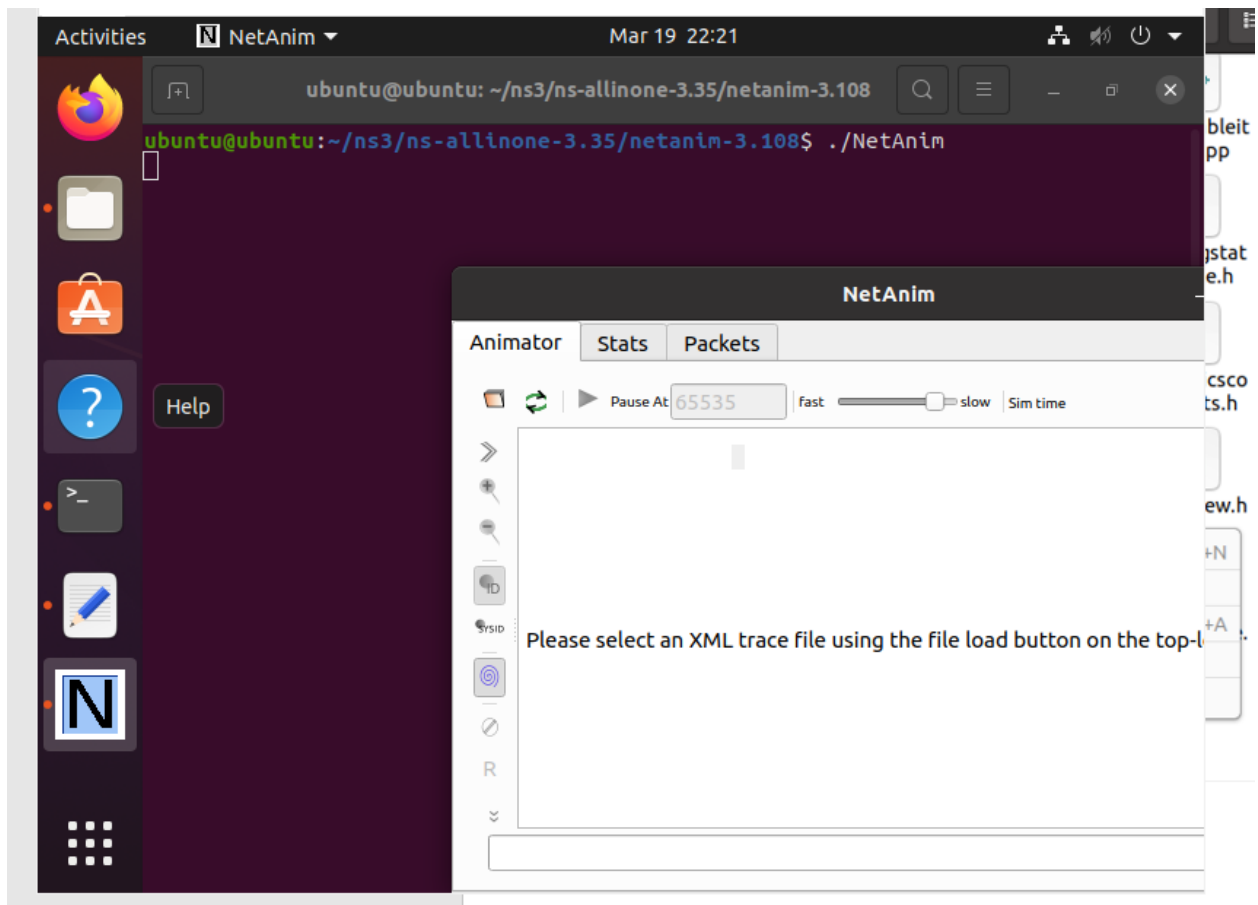
```
return 0;
```

```
}
```



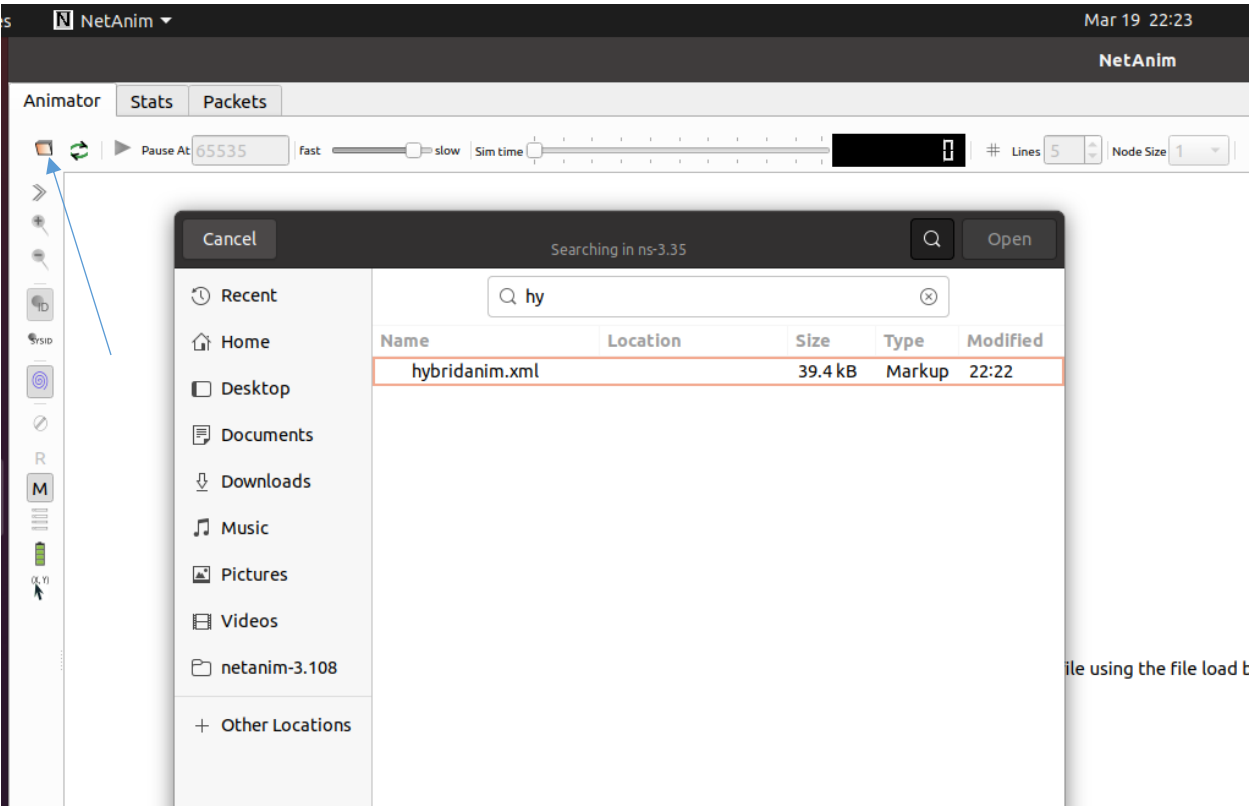


OPEN TERMINAL



```
Mar 19 22:22
ubuntu@ubuntu: ~/ns3/ns-allinnone-3.35/ns-3.35
ubuntu@ubuntu:~/ns3/ns-allinnone-3.35/ns-3.35$ ./waf --run scratch/hybridanim
Waf: Entering directory `/home/ubuntu/ns3/ns-allinnone-3.35/ns-3.35/build'
Waf: Leaving directory `/home/ubuntu/ns3/ns-allinnone-3.35/ns-3.35/build'
Build commands will be stored in build/compile_commands.json
'build' finished successfully (0.606s)
AnimationInterface WARNING:Node:1 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:2 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:3 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:1 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:2 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:3 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
At time +2s client sent 1024 bytes to 10.1.2.4 port 9
AnimationInterface WARNING:Node:2 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:3 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:2 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:3 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:2 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:3 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
At time +2.01799s server received 1024 bytes from 10.1.3.3 port 49153
At time +2.01799s server sent 1024 bytes to 10.1.3.3 port 49153
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:2 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:3 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:2 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:3 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:4 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:2 Does not have a mobility model. Use SetConstantPosition if it is stationary
AnimationInterface WARNING:Node:3 Does not have a mobility model. Use SetConstantPosition if it is stationary
At time +2.03367s client received 1024 bytes from 10.1.2.4 port 9
ubuntu@ubuntu:~/ns3/ns-allinnone-3.35/ns-3.35$
```

IN NEtAnim



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